ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE I

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Capacitors and time constants, half- and full-wave rectifiers and displays, electronic cam profiles, PWM controllers and jog/run control, hydraulic pumps and a clamp-and-press circuit with a sequence valve, electro-hydraulic continuous reciprocation, and fault-finding on drifting cylinders and bleeding air.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Write neatly and legibly.

QUESTION 1: ELECTRONICS

1.1 Name TWO types of capacitors and give ONE use of a capacitor in a mechatronic circuit. (3)

1.2 A 100 kΩ resistor is connected in series with a 47 µF capacitor. Calculate the time constant, and the time for the capacitor to be (almost) fully charged. (4)

1.3 Explain the difference between half-wave and full-wave rectification. Draw the output waveform of each. (6)

1.4 Explain how a seven-segment display shows numbers. (3)

1.5 Explain why a heat sink is fitted to a power transistor or voltage regulator. (2)

1.6 Define the term period of a waveform. (2)

[20]

QUESTION 2: DRIVES AND CONTROL

2.1 Explain the function of electronic cam profiles in a motion-control system. (3)

2.2 Name the main parts of a PWM motor controller. (3)

2.3 Give THREE methods of controlling the speed of a DC motor. (3)

2.4 Design a relay control circuit for a motor that can run continuously (RUN button, latched) or be inched (JOG button, runs only while held), with a STOP button and an emergency stop. (8)

2.5 Define the term electric drive. (3)

[20]

QUESTION 3: HYDRAULICS

3.1 Name THREE types of hydraulic pumps. (3)

3.2 Explain how an external gear pump moves oil. (4)

3.3 Draw the symbol of EACH of the following: (5)

3.3.1 Hydraulic motor (one direction) (2)

3.3.2 Pressure sequence valve (3)

3.4 Design a hydraulic circuit in which cylinder A clamps a workpiece first, and cylinder B (the press) extends only when the clamping pressure reaches a set value. Use a 4/3-way manually operated valve, a pressure relief valve and a sequence valve. (8)

[20]

QUESTION 4: ELECTRO-HYDRAULIC CONTROL

4.1 Explain the difference between a PNP and an NPN three-wire proximity sensor. (4)

4.2 Design the electrical control circuit for a double-acting cylinder with a 4/2-way double-solenoid valve (Y1 extends, Y2 retracts) that moves continuously in and out between limit switch S1 (retracted) and limit switch S2 (extended) after START is pressed, until STOP is pressed. Include an emergency stop. (12)

4.3 Give FOUR reasons why electro-hydraulic systems are used in industry. (4)

[20]

QUESTION 5: FAULT-FINDING AND MAINTENANCE

5.1 A hydraulic cylinder slowly moves down (drifts) while holding a load. Give THREE possible causes. (3)

5.2 Name FIVE things to check when inspecting a hydraulic hose. (5)

5.3 Explain how air is bled from a hydraulic system after a component has been replaced. (4)

5.4 Draw up a simple fault-finding table with the columns FAULT, POSSIBLE CAUSE and REMEDY, with TWO rows for a hydraulic system. (6)

5.5 Why must the correct type of hydraulic oil be used when topping up? (2)

[20]

TOTAL: 100

END OF QUESTION PAPER